

BAULKHAM HILLS HIGH SCHOOL MARKING COVER SHEET



YEAR 12 EXT 1
JUNE 2008

STUDENTS NAME: _____
TEACHERS NAME: _____

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	
PERCENTAGE	



YEAR 12 EXTENSION 1 MATHEMATICS ASSESSMENT JUNE 2008

TIME : 70 MINUTES

NAME		RESULT	
DIRECTIONS	- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work. - Use black or blue pen only (<i>not pencils</i>) to write your solutions. - No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.		
QUESTION 1.	Differentiate (a) $\sin^{-1} \frac{x^2}{2}$ (b) $\cos^{-1}(e^x)$		2 2
QUESTION 2.	Find the following : (a) $\int \frac{dx}{\sqrt{25-x^2}}$ (b) $\int_{\frac{1}{3}}^{\frac{1}{\sqrt{3}}} \frac{dx}{1+9x^2}$		2 3
QUESTION 3.	State the domain and range for: $y = 3\sin^{-1} 4x$ and hence draw a neat sketch of this function showing clearly the co-ordinates of the end points.		2 2

QUESTION 4.	The acceleration of a particle moving in a straight line is given by: $a = 2x - 3$ where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is at rest at $x = 4$. a) If the velocity of the particle is $v \text{ ms}^{-1}$ show that: $v^2 = 2(x^2 - 3x - 4)$ b) Show that the particle does not pass through the origin. c) Determine the position of the particle when $v = 10 \text{ ms}^{-1}$. Justify your answer.	3 2 2
QUESTION 5.	Without the use of a calculator, evaluate the following showing all working: a) $\cos\left(\tan^{-1}\left(\frac{-2}{3}\right)\right)$ b) $\tan\left[\sin^{-1}\frac{3}{5} + \cos^{-1}\frac{12}{13}\right]$	2 3
QUESTION 6.	Evaluate $\int_0^{\ln 2} \frac{e^x}{4 + e^{2x}} dx$ using the substitution $u = e^x$.	4
QUESTION 7.	Find $\int \cos^2 2x dx$.	2
QUESTION 8.	Consider the function $f(x) = 3x - x^3$ a) Sketch $y = f(x)$ showing the x and y intercepts and the co-ordinates of the stationary points. b) Find the largest domain containing the origin for which $f(x)$ has an inverse function $f^{-1}(x)$. c) State the domain of $f^{-1}(x)$. d) Find the gradient of the function $f^{-1}(x)$ at $x = 0$.	3 1 1 1
QUESTION 9.	The velocity $v \text{ ms}^{-1}$ of a particle moving along the x axis is given by $v^2 = 15 - 2x - x^2$, where $x \text{ m.}$ is its displacement from the origin. a) Show that this motion is SHM. b) Find the amplitude of this motion. c) Find its greatest velocity.	2 2 2

QUESTION 10.	A particle is projected with an initial velocity of 60 ms^{-1} at an angle of 45° to the horizontal. Taking $g = 10 \text{ ms}^{-1}$ find: a) Expressions for the horizontal and vertical displacements. b) The Cartesian equation of the particle. c) The greatest height reached by the particle. d) The angle that the path of the particle makes with the horizontal after 6 seconds.	4 2 3 2
	THE END	

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

1a)

$$\frac{2x}{\sqrt{1-\frac{2x}{y}}} = \frac{2x}{\sqrt{\frac{y-2x}{y}}} = \frac{2x}{\frac{\sqrt{y-2x}}{\sqrt{y}}} = \frac{2x\sqrt{y}}{\sqrt{y-2x}}$$

2

4

b)

$$\frac{-e^x}{\sqrt{1-e^{2x}}}$$

2

2) a)

$$\int \frac{dx}{\sqrt{25-x^2}}$$

$$= \sin^{-1} \frac{x}{5} + C$$

2

b)

$$\int_{\frac{1}{3}}^{\frac{1}{2}} \frac{dx}{1+x^2}$$

$$= \frac{1}{9} \int_{\frac{1}{3}}^{\frac{1}{2}} \frac{dx}{1+x^2}$$

$$= \frac{1}{9} \cdot \frac{3}{1} \left[\tan^{-1} 3x \right]_{\frac{1}{3}}^{\frac{1}{2}}$$

$$= \frac{1}{3} \left[\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right]$$

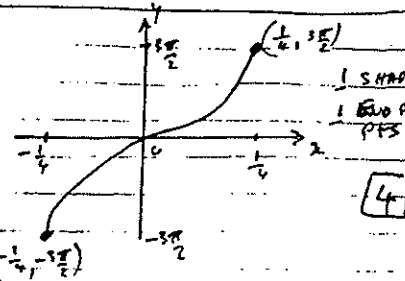
$$= \frac{1}{3} \left[\frac{\pi}{3} - \frac{\pi}{4} \right]$$

$$= \frac{\pi}{36}$$

3)

$$D \rightarrow \frac{1}{4} - \frac{1}{4} \leq x \leq \frac{1}{4} \text{ or } |x| \leq \frac{1}{4}$$

$$R \rightarrow -3\frac{\pi}{2} \leq y \leq 3\frac{\pi}{2} \text{ or } |y| \leq 3\frac{\pi}{2}$$



$$4) a) \frac{d^2y}{dx^2} = \frac{d}{dx} (2x-3) = 2$$

$$\therefore \frac{v^2}{2} = x^2 - 3x + C_1$$

$$v^2 = 2x^2 - 6x + C_2$$

$$\text{at } t=0, v=0, x=4$$

$$\therefore 32 - 24 + C_2 = 0 \therefore C_2 = -8$$

$$\therefore v^2 = 2x^2 - 6x - 8$$

$$\therefore v^2 = 2(x^2 - 3x - 4)$$

$$b) \text{ mult. } x=0$$

$$\therefore v^2 = -8$$

$$\text{no. soln}$$

$$\therefore x \neq 0$$

$$\therefore \text{does not pass thru } 0.$$

$$c) \text{ mult. } v=10$$

$$\therefore 100 = 2(x^2 - 3x - 4)$$

$$\therefore x^2 - 3x - 4 = 50$$

$$x^2 - 3x - 54 = 0$$

$$(x-9)(x+6) = 0$$

$$\therefore x=9 \text{ or } -6$$

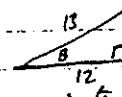
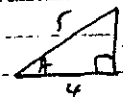
but particle starts at $x=4$ and doesn't pass thru 0

$$\therefore x \neq -6$$

$$\therefore x=9$$

7

$$5) \text{ let } \tan^{-1} \frac{3}{5} = A \quad \cos^{-1} \left(\frac{12}{13} \right) = B$$



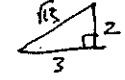
$$\therefore \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \cdot \frac{5}{12}}$$

$$= \frac{\frac{36+20}{48}}{1 - \frac{15}{48}}$$

$$= \frac{56}{33}$$

$$6) \cos \left(\tan^{-1} \left(-\frac{2}{3} \right) \right)$$



$$\frac{\pi}{2} A < 0 \text{ as } \tan^{-1} < 0$$

$$\therefore \cos \left(\tan^{-1} \left(-\frac{2}{3} \right) \right) = \frac{3}{\sqrt{13}}$$

6

$$\int_0^{\ln 2} \frac{e^x}{4+e^{2x}} dx$$

$$\text{let } u = e^x \therefore du = e^x dx$$

$$\left. \begin{matrix} x = \ln 2 & u = 2 \\ x = 0 & u = 1 \end{matrix} \right\}$$

$$= \int_1^2 \frac{du}{4+u^2}$$

$$= \frac{1}{2} \left[\tan^{-1} \frac{u}{2} \right]_1^2$$

$$= \frac{1}{2} \left[\tan^{-1} 1 - \tan^{-1} \frac{1}{2} \right]$$

$$= \frac{\pi}{8} - \frac{1}{2} \tan^{-1} \frac{1}{2}$$

7

$$\int \cos^2 x dx$$

$$\cos 4x = 2\cos^2 2x - 1$$

$$\therefore \cos^2 2x = \frac{1}{2} (1 + \cos 4x)$$

$$\therefore I = \frac{1}{2} \int (1 + \cos 4x) dx$$

$$= \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right)$$

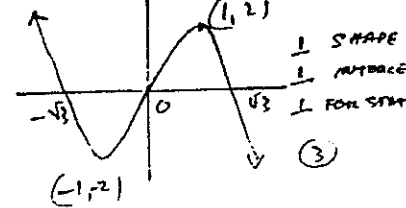
$$= \frac{x}{2} + \frac{1}{8} \sin 4x + C$$

8

$$a) f(x) = 3x - x^3$$

$$f'(x) = 3 - 3x^2 = 0 \quad x = \pm 1$$

stationary pts at (1,2) (-1,-2)
x intercepts (sqrt(3),0), 0, (-sqrt(3),0)



$$b) \text{ Domain of } f'(x) \quad -1 \leq x \leq 1$$

$$c) \text{ Domain of } f''(x) \quad -2 \leq x \leq 2$$

$$d) \frac{d}{dx} f(x) = 3 \text{ at } 0$$

$$\therefore \frac{d}{dx} f'(x) = \frac{1}{3} \text{ at } 0$$

6

① a) $v^2 = 15 - 2x - x^2$

$$\frac{v^2}{2} = \frac{15}{2} - x - \frac{1}{2}x^2$$

$$\frac{d}{dx} \left(\frac{v^2}{2} \right) = -1 - x = -1(x+1) \quad (2)$$

there is SHM about $x = -1$ $x = 1$

b) when $v = 0$

$$x^2 + 2x - 15 = 0$$

$$(x+5)(x-3) = 0 \quad (2)$$

$$x = -5 \text{ or } +3$$

$\therefore$ oscillates between -5 and $+3$
 $\therefore$ AMP = 4

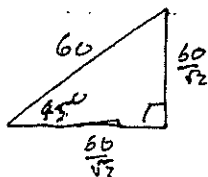
c) max velocity when $\frac{dv}{dx} = 0$ 6

$$\text{i.e. } x = -1$$

$$\therefore v^2 = 15 + 2 - 1 = 16$$

$$\therefore \text{max velocity} = 4 \text{ m s}^{-1}$$

(16)



at $t=0$ $v_y = \frac{60}{\sqrt{2}}$ $v_x = \frac{60}{\sqrt{2}}$

$$\frac{d^2y}{dt^2} = -10$$

$$\frac{dy}{dt} = C - 10t \quad C = \frac{60}{\sqrt{2}}$$

$$\frac{dy}{dt} = \frac{60}{\sqrt{2}} - 10t$$

$$y = \frac{60t}{\sqrt{2}} - 5t^2 + C$$

$$\therefore y = \frac{60t}{\sqrt{2}} - 5t^2$$

$$\frac{d^2x}{dt^2} = 0$$

$$\frac{dx}{dt} = \frac{60}{\sqrt{2}}$$

$$x = \frac{60t}{\sqrt{2}} + C \quad (4)$$

$$\text{at } x=0 \quad t=0 \quad \therefore C=0$$

$$\therefore y = \frac{60t}{\sqrt{2}} - 5t^2 \quad x = \frac{60t}{\sqrt{2}}$$

b) substit $t = \frac{\sqrt{2}x}{60}$

$$\therefore y = \frac{60}{\sqrt{2}} \cdot \frac{\sqrt{2}x}{60} - 5 \cdot \frac{2x^2}{3600}$$

$$y = x - \frac{x^2}{360} \quad (2)$$

c) max height when $\frac{dy}{dx} = 0$

$$\therefore t = \frac{6}{\sqrt{2}}$$

(3)

$$\therefore y = \frac{60}{\sqrt{2}} \cdot \frac{6}{\sqrt{2}} - 5 \cdot \frac{36}{2}$$

$$= \frac{360 - 180}{2}$$

$$= 90 \text{ m}$$

d) when $t = 6$

$$\frac{dy}{dt} = \frac{60}{\sqrt{2}} (1 - \sqrt{2}) \quad \frac{dx}{dt} = \frac{60}{\sqrt{2}}$$

$$\tan^{-1} \alpha = \left| \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \right| = \sqrt{2} - 1$$